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Subnetting Using Cisco Packet Tracer

AIM:

To implement classless IP subnetting using Cisco Packet Tracer and establish communication between multiple subnets through routers.

PROCEDURE:

1. Open Cisco Packet Tracer.
2. Create a new network topology.
3. Add routers, switches and PCs.
4. Connect the devices using appropriate cables.
5. Divide the network 192.168.1.0/24 into smaller subnets.
6. Use /27 subnet masks as specified in the manual.
7. Configure the router interfaces with suitable IP addresses.
8. Configure the switch ports.
9. Assign IP addresses and subnet masks to the PCs.
10. Configure default gateways.
11. Enable router interfaces using:
    no shutdown

Sample Output :

PC1> ping 192.168.2.11

Reply from 192.168.2.11 :
Bytes = 32 Time<1ms TTL = 126

Packets : Sent = 4
Received = 4
Lost = 0
0 % packet loss

12. Test connectivity using ping.
13. Observe packet transmission between subnets.

The manual uses /27 subnetting and provides example subnet addresses such as 192.168.1.0/27, 192.168.2.0/27, 192.168.3.0/27 and 192.168.4

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Result:

Thus, classless IP subnetting was successfully implemented in Cisco Packet Tracer and communication between the subnets was verified.